



# Change the Shape

Quick facilitation notes & answer key

## Goal

Students read and ALTER an existing loop (C3.2): change the range count and the given turn to draw a different regular shape, and describe how the shape changes. Angles are given (not derived).

## Curriculum (Ontario Math 2020, Strand C3)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

## Materials

Python Turtle projected (or a web playground); the printed worksheet + a pencil per student. A turn-angle chart (90/120/72/60) helps.

## Run it (about 30 min)

1. Start from the square; change range to 3 and the turn to 120 — run the triangle.
2. Show the shape chart; the range = number of sides, turn is given per shape.
3. Students do Exercises 1–4 (Ex 4 is the stretch), solo or in pairs.
4. Share answers; land the big idea: range = sides; the turn is given per shape.

## Answer key (checked by the run-gate)

Ex 1 — range(3), right(120).

Ex 2 — 6 sides; a hexagon.

Ex 3 — range(3).

Ex 4 (challenge) — 5 sides; a pentagon.

## Watch for & differentiate

Common slip: changing only the range and forgetting the turn must match the shape.

Support: give the turn and have students supply just the range number.

Extend: ask the range + given turn for a hexagon they have not built (range 6, right 60).

## Success looks like

The student maps a shape to its range count (Ex 1, 3), predicts the shape after altering range+turn (Ex 2), and names a polygon from a loop (Ex 4).